Total No. of Questions : 8]

SEAT No. :

PD-4889

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B.E. (Computer Engineering)

Honours in Data Science

MACHINE LEARNING AND DATA SCIENCE

(2019 Pattern) (Semester - VII) (410501)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) For what type of data Density-Based Spatial Clustering is suitable? Which parameters are required by DBSCAN algorithm. **[9]**

b) Explain KNN algorithm with example. **[8]**

OR

Q2) a) What are the types of hierarchical clustering methods? Explain. **[9]**

b) Explain K-Means algorithm with an example. **[8]**

Q3) a) How does the learning rate affect the training of the Neural Network? What do you mean by Hyperparameters? **[6]**

b) What is the use of Artificial Neural Network? How is ANN useful in making a machine intelligent? **[6]**

c) Explain back propagation algorithm. **[6]**

OR

Q4) a) Explain a biological neuron along with its parts. **[6]**

b) Explain Generalized Delta Learning Rule. **[6]**

c) What are the various components of a Neural Network? **[6]**

P.T.O.

Q5) a) Explain the terms "Valid Padding" and "Same Padding" in CNN. List down the hyperparameters of a Pooling Layer. **[6]**

b) Explain the applications where CNNs are used extensively? **[6]**

c) What are the best practices that need to be followed when designing a CNN? **[6]**

OR

Q6) a) Explain CNN Architecture along with diagram. **[6]**

b) Enlist various types of CNN models. Explain any two of them. **[6]**

c) What is the use of the convolution layer in CNN? **[6]**

Q7) a) What are various text similarity measures? Explain any two of them. **[8]**

b) Write short nonte on : **[9]**

i) Stemming

ii) Lemmatization

iii) topic modeling

OR

Q8) a) Explain the process of text processing. **[8]**

b) Write short note on : **[9]**

i) features selection

ii) tokenization

iii) document representation